Domain Adaptation for Sentiment Analysis

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Abstract

Sentiment classifiers trained on one domain frequently underperform when applied to text from a different domain, even when the underlying task is identical. This report presents a domain adaptation pipeline for cross-domain sentiment analysis that requires no labeled data from the target domain. Using IMDB movie reviews as a labeled source domain, we adapt a BERT-based model to two target domains: Amazon product reviews and Yelp business reviews through a combination of Domain-Adaptive Pre-Training (DAPT) and pseudo-labeling. DAPT continues BERT's masked language model pre-training on unlabeled target text to align the model's representations with the target domain's vocabulary and writing style, while pseudo-labeling subsequently fine-tunes the model on high-confidence predictions over unlabeled target samples. We evaluate six progressive baselines on a fully held-out test set of 10,000 samples per domain, covering a rule-based VADER baseline, zero-shot BERT transfer, DAPT alone, pseudo-labeling alone, and the full combined pipeline. The full pipeline achieves 93.79% accuracy on Amazon and 93.65% on Yelp, improving over zero-shot transfer by 2.41 and 3.27 percentage points respectively. Notably, on the Yelp domain, the unsupervised pipeline matches the performance of a model fine-tuned on 1,000 labeled target examples, demonstrating that the combination of domain pre-training and self-training can substitute for manual annotation in low-resource settings. Ablation results show that DAPT and pseudo-labeling contribute through complementary mechanisms, with their combination consistently outperforming either technique in isolation.

1. Introduction

Sentiment analysis is the task of automatically determining the sentiment expressed in a piece of text and is one of the most widely studied problems in natural language processing (NLP). Systems that perform well on this task have broad commercial and research applications, from monitoring product reviews to understanding public opinion on social media. Despite considerable progress, a persistent practical challenge is that models trained on one source of text frequently fail to generalize to a different domain, even when the underlying task is identical.

This phenomenon arises because different review platforms exhibit distinct linguistic characteristics. A movie review on IMDB tends to be lengthy and discursive, drawing on narrative conventions and cinematic vocabulary. By contrast, product reviews on Amazon mix technical terminology with everyday language and vary widely in length and register. Yelp reviews, covering local businesses such as restaurants and services, are typically shorter, more colloquial, and laden with domain-specific vocabulary relating to food quality, service, and ambience. A model trained on IMDB data acquires strong priors about movie-related sentiment cues that do not automatically transfer to these other settings.

This project addresses the cross-domain sentiment classification problem through a progressive pipeline built on BERT (Bidirectional Encoder Representations from Transformers). We use IMDB reviews as a richly labeled source domain and apply the trained model to two target domains: Amazon product reviews and Yelp business reviews. Crucially, no target-domain labels are required at training time, and the pipeline relies entirely on unlabeled target data for domain adaptation and on pseudo-labels generated automatically at inference time. The approach combines three complementary techniques:

- **Domain-Adaptive Pre-Training (DAPT)**: continued masked language model (MLM) pre-training on unlabeled target-domain text to align BERT's internal representations with the target domain.
- **IMDB Fine-Tuning**: supervised fine-tuning of the adapted model on labeled IMDB data to instill sentiment classification capability.
- **Pseudo-Labeling**: self-training on high-confidence model predictions over unlabeled target samples to further reduce the domain gap.

To quantify the contribution of each component, we evaluate six progressive baselines on a held-out test set from each target domain: a rule-based VADER baseline, BERT fine-tuned only on IMDB, a small-labeled-target oracle, pseudo-labeling without DAPT, DAPT without pseudo-labeling, and the full combined pipeline. All experiments are conducted on a fully held-out split of the respective target dataset

never seen during any training phase, and we will be using accuracy and macro-averaged F1 as the primary evaluation metrics.

2. Review of Existing Techniques/Literature Review

2.1 Lexicon-Based Sentiment Analysis

Before the advent of neural methods, sentiment analysis was dominated by lexicon-based approaches that assign sentiment scores to individual words or phrases and aggregate them to produce a document-level prediction. VADER (Valence Aware Dictionary and sEntiment Reasoner) is a widely used rule-based system designed specifically for social media and short-form text. It incorporates a human-validated lexicon of over 7,500 lexical features with associated sentiment polarity and intensity scores, and applies a set of grammatical heuristics, including handling of punctuation, capitalization, degree modifiers, and negation to produce a compound sentiment score. Despite its simplicity, VADER performs reasonably well out-of-the-box without any training, making it a natural and interpretable lower-bound baseline.

The limitation of lexicon approaches is their inability to capture context. Identical words can carry opposite sentiment in different settings, and novel or domain-specific vocabulary will not appear in any fixed lexicon. These weaknesses motivate the shift toward learned representations.

2.2 BERT and Transfer Learning

The introduction of BERT by Devlin et al. (2019) marked a turning point in NLP. BERT is a deeply bidirectional Transformer encoder pre-trained on BookCorpus and English Wikipedia using two self-supervised objectives: Masked Language Modeling (MLM), in which 15% of input tokens are masked and the model must predict them from context, and Next Sentence Prediction (NSP). The resulting representations encode rich syntactic and semantic information that can be transferred to downstream tasks through fine-tuning, adding a task-specific classification head and continuing to train on labeled examples.

For sentiment analysis, fine-tuning BERT on a labeled dataset such as the IMDB movie review corpus yields strong results within the source domain. The challenge is that this fine-tuning is domain-specific: the model learns to associate IMDB-style language patterns with sentiment labels, and these associations may not generalize when the vocabulary and writing style shift significantly, as they do between movie reviews and product or restaurant reviews.

2.3 Domain-Adaptive Pre-Training

Gururangan et al. (2020) demonstrated systematically that continuing BERT's MLM pre-training on unlabeled text from the target domain, a technique they termed Domain-Adaptive Pre-Training (DAPT), leads to consistent performance improvements on downstream tasks in that domain. The intuition is that additional exposure to domain-specific text allows the model to update its internal token embeddings and attention patterns to better reflect the statistical regularities of the target vocabulary, writing style, and syntax, even without any labeled signal.

DAPT is particularly attractive in cross-domain settings because it is entirely unsupervised: it requires only raw text from the target domain. The computational cost is moderate compared to training from scratch, since only additional pre-training steps are required rather than a full re-initialization. In our pipeline, DAPT is applied to 60% of the target domain corpus with labels stripped before any supervised fine-tuning is performed.

2.4 Pseudo-Labeling and Self-Training

Pseudo-labeling, also known as self-training, is a semi-supervised learning technique in which a model trained on labeled data is used to generate soft or hard labels for unlabeled samples. Only predictions that exceed a confidence threshold are retained as pseudo-labels; these are then used to augment the training set for a further round of fine-tuning. The key intuition is that even if the teacher model is imperfect, its high-confidence predictions are likely to be correct, and training on them can help the student model learn target-domain representations it would not otherwise encounter.

Lee (2013) popularized this approach for deep networks, and it has since been widely adopted in NLP for domain adaptation and low-resource settings. The quality of pseudo-labels depends critically on the confidence threshold: too low, and label noise degrades performance; too high, and too few samples are retained to provide any useful signals. In our experiments we use a threshold of 0.9, meaning only samples where the model assigns at least 90% probability to one class are included. Pseudo-labeling is applied to 20% of the target corpus to generate additional training signals before evaluation on the remaining held-out dataset.

3. Methods

3.1 Datasets and Data Preparation

Three datasets are used in this project. The IMDB Large Movie Review Dataset serves as the labeled source domain, providing 50,000 training examples of positive and negative movie reviews. The Amazon Product Reviews dataset and the Yelp Reviews dataset each serve as unlabeled target domains, with reviews binarized into positive and negative labels.

For each target domain, the data is split into three non-overlapping pools using stratified sampling to preserve class balance:

- **Pool A (60%, ~30,000 samples)**: used exclusively for DAPT. Labels are stripped and only raw text is retained.
- **Pool B (20%, ~10,000 samples)**: used for pseudo-label generation. Also provides 1,000 samples for the small-labeled-target oracle baseline (Baseline 3).
- **Pool C (20%, ~10,000 samples)**: the held-out test set used for all final evaluations. This split is never seen during any training phase.

3.2 Pipeline Overview

The full pipeline proceeds in four stages, each building on the previous:

Stage 1 (DAPT): The base bert-base-uncased model is loaded and its MLM pre-training is continued on the raw text of Pool A for 3 epochs with a learning rate of 2e-5 and a batch size of 16. The adapted model is saved for use in subsequent stages.

Stage 2 (IMDB Fine-Tuning): A classification head (linear layer mapping the [CLS] token representation to two output classes) is added to the DAPT-adapted model. This combined model is fine-tuned on the full 50,000-sample IMDB training set for 3 epochs with a learning rate of 2e-5.

Stage 3 (Pseudo-Labeling): The fine-tuned model from Stage 2 is used as a teacher to generate predictions over Pool B. Samples where the softmax output assigns probability ≥ 0.9 to either class are retained with their predicted label as pseudo-ground-truth. The model is then fine-tuned for a further 3 epochs on this pseudo-labeled subset.

Stage 4 (Evaluation): The final model is evaluated on Pool C using accuracy and macro-averaged F1 score. All six baselines are evaluated on the same Pool C split.

3.3 Progressive Baselines

Six baselines are evaluated to isolate the contribution of each pipeline component:

- **Baseline 1 (VADER):** Rule-based lexicon approach; no training required. Serves as an interpretable lower bound.
- **Baseline 2 (BERT + IMDB):** Vanilla BERT fine-tuned on IMDB with no domain adaptation. Establishes the zero-shot cross-domain transfer baseline.
- **Baseline 3 (BERT + IMDB + Small Labelled Target):** BERT + IMDB further fine-tuned on 1,000 labeled target samples from Pool B. Provides an approximate upper bound for what minimal supervised adaptation achieves.
- **Baseline 4 (BERT + IMDB + Pseudo):** Pseudo-labeling applied without prior DAPT. Isolates the effect of pseudo-labeling independently of domain pre-training.
- **Baseline 5 (BERT + DAPT + IMDB):** DAPT followed by IMDB fine-tuning but without pseudo-labeling. Isolates the effect of domain pre-training.
- **Baseline 6/Full Pipeline (BERT + DAPT + IMDB + Pseudo):** All components combined. This is the primary proposed method.

3.4 Implementation Details

All experiments use the bert-base-uncased checkpoint from Hugging Face Transformers as the base model. Training is performed on NVIDIA T4 GPUs via Google Colab. The HuggingFace Trainer API is used with the following shared hyperparameters: learning rate 2e-5, AdamW optimizer, weight decay 0.01, linear learning rate warmup over the first 10% of steps, and batch size 16. Evaluation uses macro-averaged F1 as the primary metric to account for any class imbalance in the test sets.

4. Experiments and Results

4.1 Amazon Reviews

Table 1 reports the accuracy and macro F1 scores for all six baselines on the held-out Amazon Pool C test set (10,000 samples).

Baseline	Accuracy	Macro F1
1. VADER	0.7247	0.7128
2. BERT + IMDB	0.9138	0.9136
3. BERT + IMDB + Small Labelled Amazon	0.9440	0.9440
4. BERT + IMDB + Pseudo (no DAPT)	0.9321	0.9321
5. BERT + DAPT + IMDB	0.9317	0.9316
6. BERT + DAPT + IMDB + Pseudo (Full)	0.9379	0.9378

Table 1: Results on held-out Amazon Pool C (10,000 samples)

VADER achieves an accuracy of 72.47% and macro F1 of 0.7128, confirming that a purely lexicon-based approach is substantially weaker than any learned method. Vanilla BERT fine-tuned on IMDB (Baseline 2) achieves 91.38% accuracy, demonstrating that BERT's pre-trained representations provide strong cross-domain transfer even without explicit domain adaptation. This is a notable result: despite no exposure to Amazon text during training, the model correctly classifies more than 9 in 10 reviews.

The small-labeled-target oracle (Baseline 3), which uses only 1,000 Amazon labels for additional fine-tuning, achieves 94.40% accuracy, the highest of all baselines and a natural ceiling for what minimal

supervision can achieve. Both pseudo-labeling alone (Baseline 4: 93.21%) and DAPT alone (Baseline 5: 93.17%) improve substantially over the no-adaptation baseline, and their contributions are approximately additive: the full pipeline (Baseline 6: 93.79%) achieves the best unsupervised result, approaching within 0.6 percentage points of the supervised oracle.

4.2 Yelp Reviews

Table 2 reports the corresponding results on the held-out Yelp Pool C test set (10,000 samples).

Baseline	Accuracy	Macro F1
1. VADER	0.7199	0.7015
2. BERT + IMDB	0.9038	0.9037
3. BERT + IMDB + Small Labelled Yelp	0.9363	0.9363
4. BERT + IMDB + Pseudo (no DAPT)	0.9161	0.9160
5. BERT + DAPT + IMDB	0.9279	0.9278
6. BERT + DAPT + IMDB + Pseudo (Full)	0.9365	0.9365

Table 2: Results on held-out Yelp Pool C (10,000 samples)

The pattern on Yelp broadly mirrors Amazon, but with one striking difference: the full pipeline (Baseline 6: 93.65% accuracy, 0.9365 F1) essentially matches the small-labeled-target oracle (Baseline 3: 93.63%, 0.9363 F1), despite using zero Yelp labels. This result demonstrates that the combination of DAPT and pseudo-labeling can, in some cases, fully replicate the benefit of a small labeled target set without any human annotation.

As with Amazon, both DAPT and pseudo-labeling individually improve over the no-adaptation baseline. However, pseudo-labeling without DAPT (Baseline 4: 91.61%) shows a smaller absolute gain on Yelp than on Amazon, while DAPT alone (Baseline 5: 92.79%) shows a larger relative contribution. This suggests that Yelp's vocabulary and writing style diverge more from IMDB than Amazon's do, making domain pre-training proportionally more valuable for the Yelp domain.

4.3 Cross-Domain Comparison

Comparing the two target domains reveals that zero-shot transfer from IMDB is marginally stronger for Amazon (91.38%) than for Yelp (90.38%), consistent with Amazon and IMDB reviews sharing a more similar writing register, whereby both tend toward longer, more considered evaluations as compared to Yelp reviews, which are shorter, more casual, and more location-specific. This difference in domain proximity is also reflected in the fact that DAPT provides a larger absolute gain on Yelp (+2.41%) than on Amazon (+1.79%): the greater the initial domain gap, the more useful continued pre-training on target-domain text becomes.

5. Discussion

5.1 Contribution of Each Component

The progressive baseline design allows clean attribution of performance gains to individual components. Moving from Baseline 2 (BERT + IMDB) to Baseline 5 (BERT + DAPT + IMDB) adds approximately 1.8 percentage points on Amazon and 2.4 percentage points on Yelp. This demonstrates that DAPT provides a meaningful but moderate benefit: it adapts BERT's token representations to target-domain vocabulary and syntax, reducing the effective domain gap that the classification head must overcome.

Pseudo-labeling (comparing Baseline 2 to Baseline 4) provides a similar magnitude of gain (approximately 1.8% on Amazon and 1.2% on Yelp), suggesting it contributes through a complementary mechanism: rather than changing the model's internal representations, it provides additional labeled-style training signal in the target domain. The full pipeline, which combines both, yields the best unsupervised performance on both domains, consistent with the two techniques being largely orthogonal in what they improve.

5.2 Comparison to the Supervised Oracle

On Amazon, the full pipeline falls 0.61% short of the oracle (Baseline 3), while on Yelp it essentially matches it. This is a practically significant result: the pipeline requires no manual annotation of target-domain data, yet achieves accuracy within a fraction of a percent of a model that has seen 1,000 labeled Yelp examples. For settings where annotation is expensive or infeasible, this represents a compelling alternative.

The fact that the oracle gap is larger on Amazon than Yelp may seem counterintuitive given that IMDB-to-Amazon transfer is generally stronger. One possible explanation is that Amazon reviews exhibit greater product-category diversity than Yelp reviews, meaning that 1,000 labeled Amazon samples provide more disambiguating information across heterogeneous subdomains, information that neither DAPT nor pseudo-labeling can fully replicate from unlabeled data alone.

5.3 Domain Proximity and Domain Gap

The empirical results provide consistent evidence that Amazon is a closer domain to IMDB than Yelp. Zero-shot transfer accuracy is approximately one percentage point higher on Amazon, and the gain from DAPT is smaller. Qualitatively, this is plausible: both IMDB and Amazon reviews tend to be longer and more evaluative in style, while Yelp reviews are shorter, more conversational, and draw heavily on location-specific vocabulary that rarely appears in either movie or product reviews.

This domain proximity argument has practical implications for dataset selection: when choosing a source domain for cross-domain transfer, a domain with similar linguistic characteristics to the target will generally yield better zero-shot performance and require less adaptation effort.

5.4 Limitations

Several limitations of the current work are worth acknowledging. First, DAPT was conducted for only 3 epochs on approximately 30,000 samples per target domain, which may be insufficient to fully adapt BERT's representations, particularly for a domain as distinctive as Yelp. Longer or more data-intensive DAPT may yield additional gains. Second, the quality of pseudo-labels was not directly measured; while the 0.9 confidence threshold was chosen to minimize label noise, some incorrectly labeled samples will inevitably have been included, and the sensitivity of results to this threshold was not systematically explored. Third, all experiments use binary sentiment classification; the methods may behave differently under multi-class or fine-grained sentiment settings. Finally, only a single source domain (IMDB) was tested; the relative contributions of DAPT and pseudo-labeling may differ when the source domain is more or less similar to the target.

5.5 Future Work

Several directions could extend this work. More principled pseudo-label selection: for example, using entropy-based filtering or consistency regularization may reduce label noise and improve gains from self-training. Combining multiple source domains (e.g., IMDB and Amazon together) when adapting to Yelp could provide a richer starting point. Finally, more recent pre-trained language models such as RoBERTa or domain-specific models like review-trained BERT variants could serve as stronger backbones, potentially amplifying the gains from adaptation.

6. Conclusion

This project presented a domain adaptation pipeline for sentiment analysis that combines Domain-Adaptive Pre-Training, supervised fine-tuning on IMDB, and pseudo-labeling to transfer a BERT-based model to unlabeled target domains without any target-domain labels. Experiments on Amazon and Yelp reviews demonstrated consistent improvements over both a lexicon-based baseline and a zero-shot BERT transfer baseline, with the full pipeline achieving 93.79% accuracy on Amazon and 93.65% on Yelp. On the Yelp domain, the pipeline matched the performance of a model trained with 1,000 labeled target examples, highlighting the practical value of the approach for annotation-scarce settings. The progressive ablation design clearly showed that DAPT and pseudo-labeling contribute through complementary mechanisms and that their combination is more effective than either alone.

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